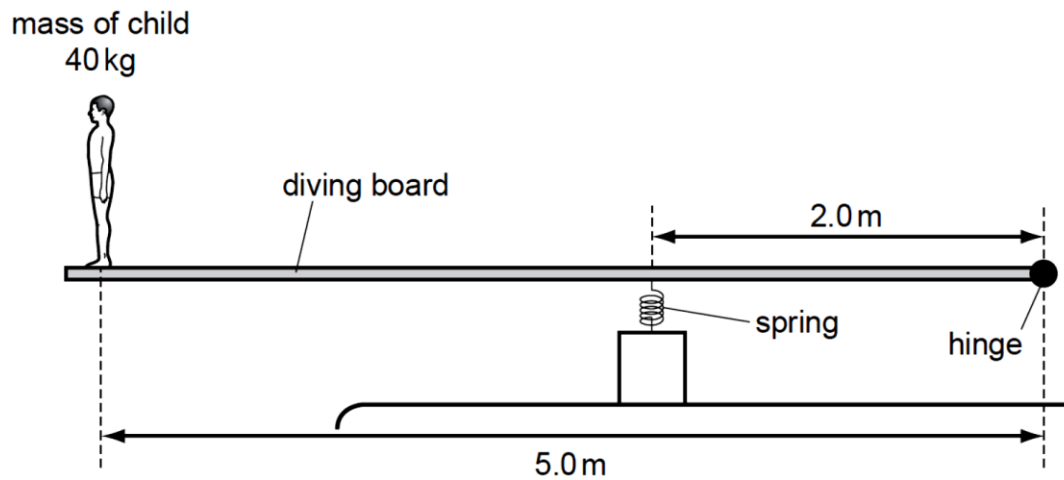


- 7 A diving board of length 5.0 m is hinged at one end and supported 2.0 m from this end by a spring of spring constant 10 kN m^{-1} . A child of mass 40 kg stands at the far end of the board.



What is the compression of the spring caused by the child standing on the end of the board?

- | | | | | | | | |
|----------|--------|----------|--------|----------|--------|----------|-------|
| A | 1.0 cm | B | 1.6 cm | C | 9.8 cm | D | 16 cm |
|----------|--------|----------|--------|----------|--------|----------|-------|

Answer: C

Taking moments about hinge,
Sum of anticlockwise moments = Sum of clockwise moments
 $5.0(40 \times 9.81) = 2.0 F$
 $5.0(40 \times 9.81) = 2.0 (10\,000)x$
 $x = 0.0981 \text{ m} = 9.81 \text{ cm} = 9.8 \text{ cm}$

where $F = kx$

- 8 The diagram shows the velocity-time graph of a 500 kg elevator during its upward motion